

Welcome to the “Beyond Arduino” workshop!

I’m immensely excited you’ve chosen to attend and I’m incredibly grateful to be given the opportunity to present this workshop to you! With any luck, by the end of the workshop you’ll not only be programming an STM32 microcontroller to do fun and useful things but you’ll also have the skills to do that on your own with any other microcontroller you wish to use in the future.

Before you arrive, it’s important that you:

- watch [this video](#) (presentation materials are available [here](#)) about how microcontrollers are programmed,
- download and install [STM32CubeIDE](#) (you’ll need to sign up for a free “MyST” account in order to do this), and
- (optionally) download and install a serial connection program like [PuTTY](#) or [Teraterm](#) (STM32CubeIDE has one built in, but it has some oddities that make it a little difficult to use).

If you wish to get the most out of this workshop, you can also:

- watch [this video](#) from Shawn Hymel about getting started with STM32 microcontrollers (it’s basically a 15-minute summary of what we’ll do in the first half of our workshop!);
- start the [“Scavenger Hunt”](#), the first exercise we’ll complete in the workshop;
- ensure you know how to use C [structs](#) and [pointers](#) to create and use [objects](#) (the STM32 HAL makes heavy use of these “objects” to allow the programmer to more easily use the microcontroller); and
- learn or review how the [UART](#) and [I2C](#) communication protocols work, since we’ll use them in our workshop.

On the day of the workshop, please bring with you:

- a laptop computer with charging cable, and
- your preferred note-taking material (if separate from your laptop).

You may also find it useful to bring:

- headphones (for watching instructional videos while you’re working on your projects) and
- a small USB logic analyzer or mixed-signal USB oscilloscope (like from Salae or Digilent).

See you all at the conference!

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Nathan Jones